

PRAMOD KUMAR MODEPU

PERSONAL DETAILS

 Hyderabad

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SKILLS

- Python
- SQL
- OOPS Concepts
- TensorFlow
- PyTorch
- Hugging Face
- Lang Chain
- AWS
- S3
- EC2
- Lambda
- GCP
- Cloud Run
- Vertex AI
- Django
- Flask
- HTML
- CSS
- JavaScript
- Git
- PyCharm
- Jupyter Notebook
- Verification and validation tools for Python
- MySQL
- Oracle
- Databricks
- Linux/Unix
- Windows

ACHIEVEMENTS

Received OPERATIONAL EXCELLENCE award for consistent and excellent performance focusing on client results

LANGUAGES

English:

Telugu:

Hindi:

DISCLAIMER

I hereby declare that all the information provided is true to the best of my

Software Engineer with 7 years of experience in Python development and 4 years of specialized expertise in Artificial Intelligence (AI), Machine Learning (ML), and Large Language Models (LLMs). Proven ability to design, build, and scale backend systems and AI-powered applications using Python, Django, FastAPI, and modern ML frameworks.

Hands-on experience in deploying and fine-tuning LLMs using LangChain for Retrieval-Augmented Generation (RAG) with both structured (Databricks, SQL) and unstructured data (Blob Storage). Developed and implemented NLP solutions including sentiment analysis, text summarization, and document classification using OpenAI APIs, Hugging Face models, and deep learning frameworks like TensorFlow and PyTorch.

Extensive work on scalable cloud infrastructure using AWS (EC2, Lambda, S3, CloudFormation) and GCP (Cloud Run, Vertex AI) for AI/ML model deployment. Strong command of RESTful API development, object-oriented design, modular architecture, CI/CD integration, and Test-Driven Development (TDD). Known for delivering production-ready applications with performance and maintainability in mind.

EXPERIENCE

March 2021 - Current

Senior Developer *EagleNet LLC*.

- Designed, developed, and deployed AI/ML use cases using Python, TensorFlow, and PyTorch.
- Built robust and scalable systems leveraging AWS for deployments and operational efficiency.
- Conducted data analysis and implemented SQL-based data solutions for insights and reporting.
- Collaborated with cross-functional teams to translate business requirements into technical solutions.
- Developed and maintained web applications using Django and RESTful APIs.
- Contributed to the continuous improvement of AI/ML infrastructure and deployment processes.

August 2018 - March 2021

Python Django Developer *HDFC ERGO*

- Led the development of motor insurance SaaS applications, implementing premium calculation and policy renewal features.
- Designed scalable back-end solutions and API endpoints for enterprise applications.
- Enhanced application performance and user experience through optimization.

EDUCATION

Bachelor of Technology Computer Science & Engineering
JNTUH University

PROJECTS

Project: RAG-based Large Dataset Query System

Mar 2024 – Present

- Built a **Retrieval-Augmented Generation (RAG)** pipeline leveraging LangChain to process and retrieve insights from large-

knowledge.

scale structured and unstructured datasets.

- Integrated with **vector databases** to enable semantic search capabilities and high-recall information retrieval for LLMs.
- Enabled seamless connection with LLMs to parse and analyze massive datasets for intelligent querying.
- Ensured robustness and scalability to handle large query loads and data volumes.

Project: LLM Orchestration on Azure Fabric with Retry Logic

Apr 2024 – Present

- Designed and implemented a system to connect **Large Language Models (LLMs)** with **Azure Fabric** for task execution across distributed services.
- Developed a retry mechanism with **3 execution trails per failed case** to ensure fault tolerance and system reliability.
- Led **schema migration** processes to maintain data model integrity during continuous deployment cycles.
- Improved orchestration efficiency and execution success rates in dynamic environments.

Mar 2021 – Aug 2022 | AI/ML Applications

- Designed and deployed AI models using **TensorFlow** and **Vertex AI**
- Built scalable data pipelines for structured and unstructured data
- Worked with cross-functional teams to convert business needs into AI/ML use cases
- Used **GCP Cloud Run** to automate model training and deployment workflows
- Tuned model performance, improving prediction accuracy by **25%**
- Created documentation for easy maintenance and updates

Mar 2021 – Aug 2022 | Content Moderation & Compliance

- Built automated moderation system using **Hugging Face** models for text/image content
- Developed **AWS Lambda** functions for on-demand moderation of S3 uploads
- Implemented real-time content flagging to reduce review time by **40%**
- Integrated external APIs for alerts and report generation
- Fine-tuned models to improve detection accuracy by **30%**
- Scaled system to handle over **1 million items/month** with zero downtime

Jan 2021 – Dec 2023 | OTT, Subscription, AI Chatbots

- Implemented **RevenueCat** REST APIs for subscription workflows on Android/iOS
- Built AI chatbots for user assistance and personalized recommendations
- Used **TensorFlow** and **PyTorch** for advanced content recommendation
- Created SQL pipelines to track and predict user behavior
- Developed predictive models with data scientists, improving retention by **25%**

- Optimized API and data fetching, boosting performance by **30%**
- Ensured app stability with **99% crash-free rate** across devices

Jan 2023 – Present | Digital Signage, OTT, Casting, Moderation

- Deployed AI/ML models to moderate hate speech and inappropriate content
- Built APIs for IoT casting with real-time updates
- Used **AWS Lambda** for scalable serverless architecture
- Implemented **TensorFlow** for recommendation and content tagging
- Created secure **HLS streaming** for global video delivery
- Integrated **Stripe** for subscription billing at scale
- Built real-time AI-powered dashboards to track user and content metrics

Jan 2017 – Dec 2021 | Insurance

- Developed a B2B platform for motor policy management using **Django & MySQL**

- Automated renewal notifications via email, SMS, and in-app messages (↑retention by 20%)
- Integrated secure payment gateways for purchases and renewals
- Built real-time dashboards for policy insights and customer engagement
- Scaled backend to support **100K+ policies** with low latency
- Reduced operational costs via workflow automation with **Celery**